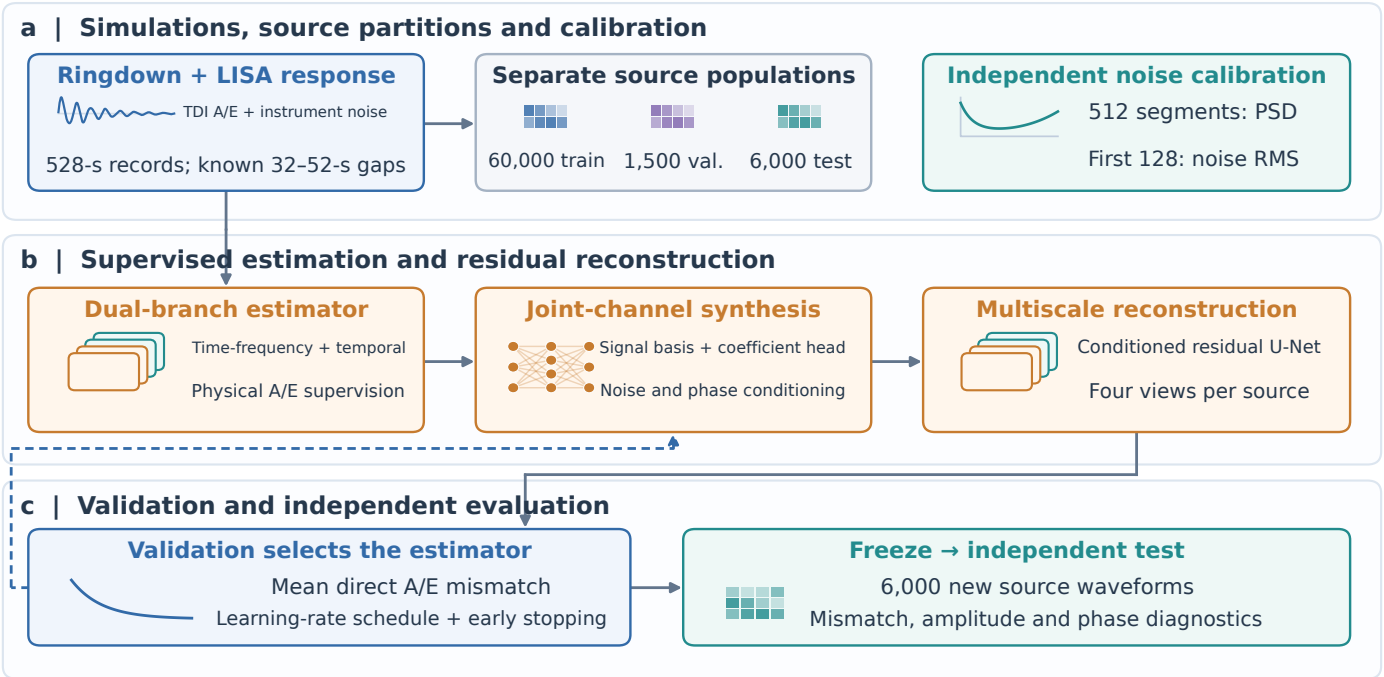
a | Simulations, source partitions and calibration
Ringdown + LISA response
TDI A/E + instrument noise
528-s records; known 32–52-s gaps
Separate source populations
60,000 train
1,500 val.
6,000 test
Independent noise calibration
512 segments: PSD
First 128: noise RMS
b | Supervised estimation and residual reconstruction
Dual-branch estimator
Time-frequency + temporal
Physical A/E supervision
Joint-channel synthesis
Signal basis + coefficient head
Noise and phase conditioning
Multiscale reconstruction
Conditioned residual U-Net
Four views per source
c | Validation and independent evaluation
Validation selects the estimator
Mean direct A/E mismatch
Learning-rate schedule + early stopping
Freeze → independent test
6,000 new source waveforms
Mismatch, amplitude and phase diagnostics